

UD65640A

CMOS Backplane for 0.25”
Micro-OLED

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1. General Descriptions

The UD65640 is a 0.25 inch CMOS backplane for micro OLED panel with resolution of 640 (RGB) × 480 dots. It integrates the complete pixel circuit, pixel driver, timing controller, and analog/digital circuits for the portable display solution. It also includes the gamma correction, deburn-in, temperature sensor, and temperature compensation necessary for OLED adjustment. To save overall power consumption, pixel memories are embedded in the backplane, which makes this backplane an excellent solution for portable applications.

2. Features

- 0.25" (Max) active display area with 3256 pixels per inch (PPI)
- Display resolution options:
 - 640 (RGB) x 480, 640 (RGB) x 400 Color display
 - 640 x 480, 640 x 400 Monochrome display
- Interface options:
 - MIPI + I2C
 - MIPI DSI (1 clock and 2 data lanes), with maximum data rate of 500Mbps per lane.

Supports both video and command modes for display data.
- Embedded memory to support dynamic refresh, i.e., no refresh is needed if display data is not changed to save panel power.
- Supports 24-bit full-color mode with 16.7 million colors.
- Scan direction selection: Select up or down scan directions.
- Display directions: Select left or right display directions.
- Orbit function supported: Configurable ±6 pixels vertical/±4 pixels horizontal display position adjustments.
- Deburn-In function supported.
- Embedded temperature compensation function.
- Internal OTP: For gamma data and trim data storage.
- Operating conditions:
 - IOVCC: typical 1.8V or 3.3V (for Interface Power)
 - AVCI18: typical 1.8V (for DVDD Regulator/Analog Power Supply)

- AVCI33: typical 3.3V (for Analog Power Supply)
- Operating temperature: TBD.

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3. System Block Diagram

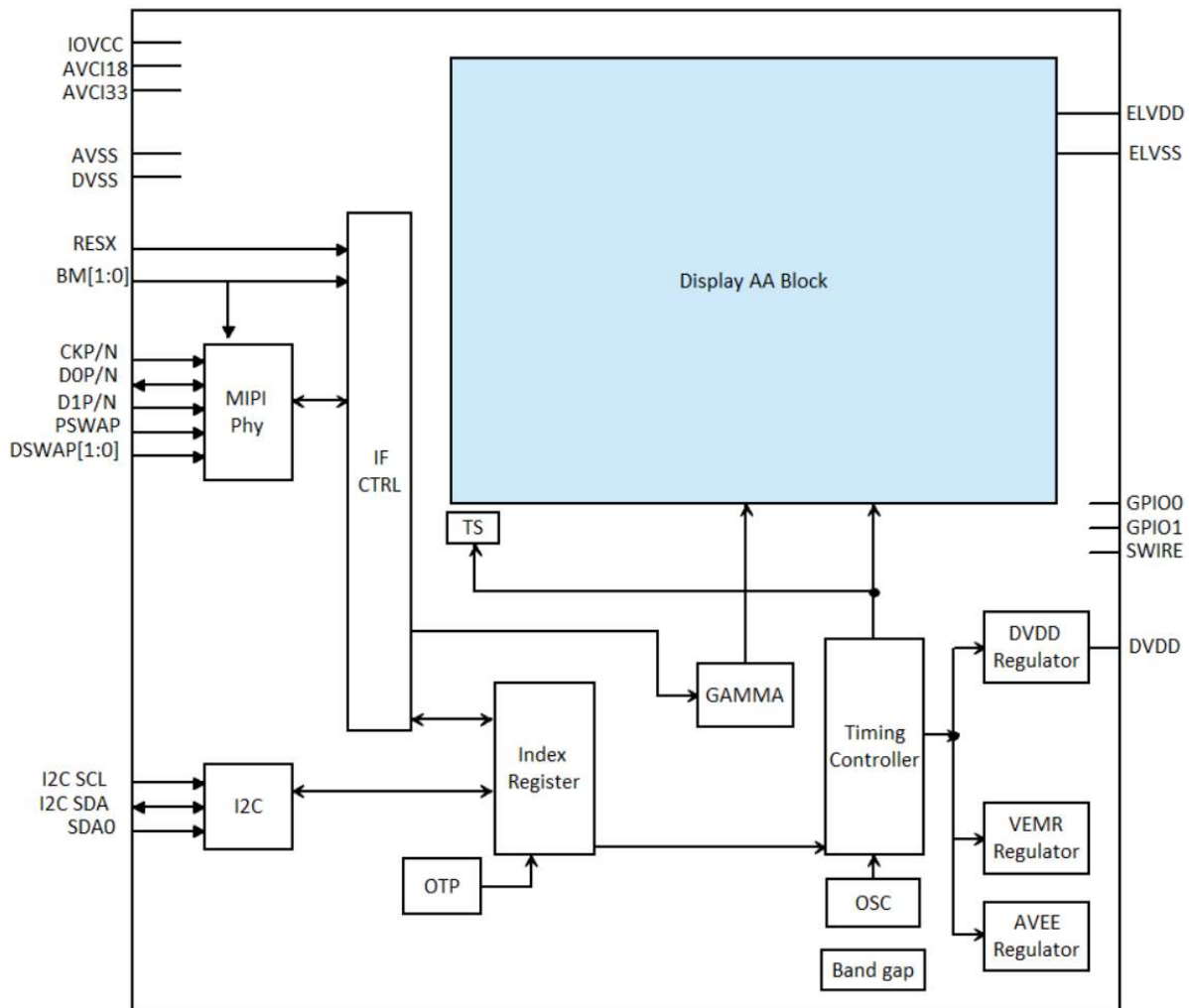


Figure 1 - System Block Diagram

4. Pin Descriptions

4.1 Pin Arrangement



Figure 2 - Pin Arrangement (Top View)

4.2 Pin Descriptions

This description applies to the MIPI interface. For other interfaces, please refer to the relevant application note.

Pin No.	Symbol	Type	Pin Description	When not used
1	NTEST	Output	Analog test pin, leave open.	Open
2	VPP	Power Supply	OTP programming power, leave open when not used.	Open
3	ELVDD	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	NA
4	ELVSS	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	NA
5	IOVCC	Power Supply	1.8V or 3.3V Power supply for interface system except MIPI interface.	NA
6	GPIO1	Input/Output	Multi-function pin, general purpose I/O. Refer to the application circuit for detailed usage.	Open
7	Reserved	Input	Do not use, leave open.	Open
8	Reserved	Input	Do not use, leave open.	Open
9	Reserved	Input	Do not use, leave open.	Open
10	Reserved	Input	Do not use, leave open.	Open
11	Reserved	Input	Do not use, leave open.	Open
12	Reserved	Input	Do not use, leave open.	Open
13	Reserved	Output	Do not use, leave open.	Open
14	Reserved	Output	Do not use, leave open.	Open
15	Reserved	Input	Do not use, leave open.	Open
16	Reserved	Input	Do not use, leave open.	Open
17	Reserved	Input	Do not use, leave open.	Open
18	Reserved	Input	Do not use, leave open.	Open

19	BIST_SEL1	Input	Build-in self-test patten select, Test pin, leave open.	Open
20	BIST_SEL0	Input	Build-in self-test patten select, Test pin, leave open.	Open
21	BIST_EN	Input	Build-in self-test enable, Test pin, leave open.	Open
22	BM1	Input	Interface type selection pin. Refer to the application circuit for details.	DVSS/Open
23	TE	Output	Tearing effect output, signal to synchronize driver, leave open when not used.	Open
24	GPIO0	Input/Output	Multi-function pin, general purpose I/O. Refer to the application circuit for detailed usage.	Open
25	DSWAP1	Input	MIPI data lane swap control pin.	DVSS/Open
26	DSWAP0	Input	MIPI data lane swap control pin.	DVSS/Open
27	PNSWAP	Input	MIPI lane polarity swap control pin.	DVSS/Open
28	D0P	Input	D0P for MIPI I/F. Refer to the application circuit for details.	Open
29	D0N	Input	D0N for MIPI I/F. Refer to the application circuit for details.	Open
30	CKP	Input	CKP for MIPI I/F. Refer to the application circuit for details.	Open
31	CKN	Input	CKN for MIPI I/F. Refer to the application circuit for details.	Open
32	D1P	Input	D1P for MIPI I/F. Refer to the application circuit for details.	Open
33	D1N	Input	D1N for MIPI I/F. Refer to the application circuit for details.	Open
34	SWIRE	Output	SWIRE output pin used to control the power IC (ELVSS power). SWIRE Pulse can from Register Programmable	Open
35	TESTE	Input	Test pin, leave open.	Open
36	ATEST	Output	Analog test pin, leave open.	Open
37	AVCI33	Power Supply	3.3V Power supply for analog circuit, connect a capacitor for stabilization.	N/A
38	AVSS	Power Supply	Analog ground.	N/A
39	DVSS	Power Supply	Digital ground.	N/A
40	DVDD	Power Supply	Regulator output to generate voltage for internal logic, connect a capacitor for stabilization.	N/A
41	AVCI18	Power Supply	1.8V Power supply for analog circuit, connect a capacitor for stabilization.	N/A
42	RESX	Input	Active-low reset pin.	N/A
43	SDA0	Input	Pin used to set the I2C device address.	DVSS/IOVCC
44	SDA	Input/Output	Bi-directional data pin for the I2C interface. Connect to IOVCC if not used.	IOVCC
45	SCL	Input	Synchronous clock signal pin for the I2C interface. Connect to IOVCC if not used.	IOVCC
46	AVCI18	Power Supply	1.8V Power supply for analog circuit, connect a capacitor for stabilization.	N/A
47	ELVSS	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	N/A
48	ELVDD	Power Supply	Power supply for OLED cell, connect a capacitor for stabilization.	N/A
49	BM0	Input	Interface type selection pin. Refer to the application circuit for details.	DVSS/Open

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50	Reserved	Input	Leave open.	Open
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Table 1 – Pin Description

5. Application Example

5.1 MIPI DSI with I2C Example

Regarding power supply capacitor connections, mount an approximately 4.7μF to 1μF capacitor for each power supply. Insufficient capacitance may affect the picture quality.

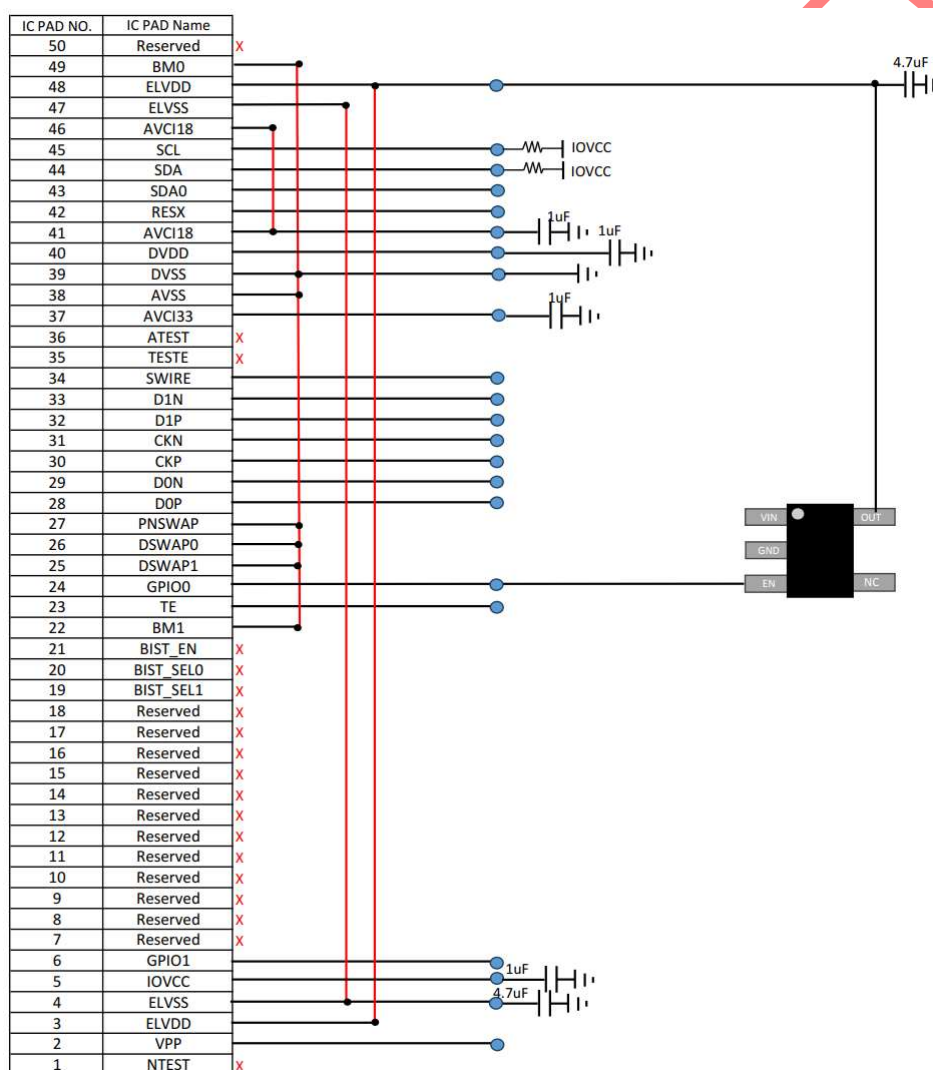


Figure 3 – MIPI DSI with I2C Example Circuit

Application Note:

No.	Pin Name	Typical Value	Max. Rated Voltage	Note
1	AVCI33	Cap, 1.0uF	4V	
2	AVCI18	Cap, 1.0uF	4V	

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3	IOVCC	Cap, 1.0uF	4V	
4	DVDD	Cap, 1.0uF	4V	
5	ELVDD should connect together at PCB to external power supply with a stabilization capacitor.	Cap, 4.7uF	4V	
6	ELVSS should connect together at PCB to external power supply with a stabilization capacitor.	Cap, 4.7uF	10V	

Table 2 – External Component

6. Interface

BM[1:0] are used to select command/image interface for UD65640.

When BM[1:0]=00, the MIPI DSI interface is selected. Image data transfer uses the 24-bit (full-color) video/command mode, while commands could be issued either via I2C or MIPI DSI. Please note that, in this mode, the I2C SCL/SDA pins should be externally pulled up, whether used or not.

6.1 I2C Interface

UD65640 functions as a slave device on the I2C bus. Whenever the I2C mode is selected or enabled, both lines must be connected to the IOVCC via pull-up resistors. UD65640 supports two device addresses. When the SDA0 pin is connected to DVSS, the device address is set to 6Ah. Alternatively, if SDA0 is tied to IOVCC, the device address becomes 6Bh. It is essential to establish a fixed SDA0 signal level before initiating any I2C communication.

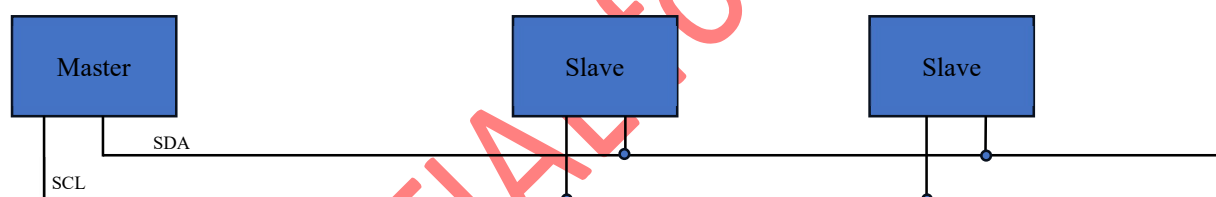


Figure 4 – I2C Connections

6.1.1 Write Command Sequence

UD65640 supports command write sequence via I2C-bus transfer. Detail transfer sequences are below.

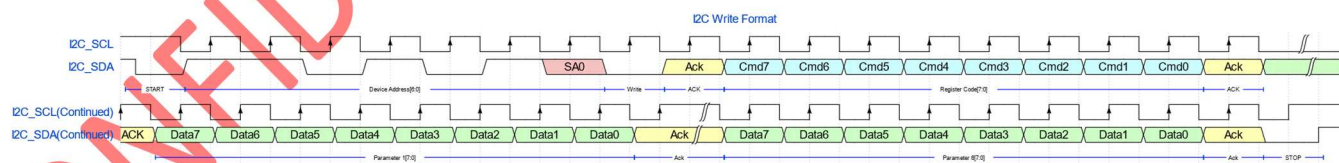


Figure 5 - Example of I2C Write Cycle Sequence

6.1.2 Read Command Sequence

The command reading transfers are shown below.

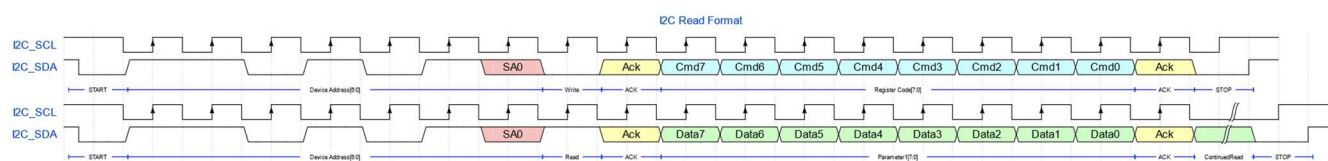


Figure 6 - Example of I2C Read Cycle Sequence

6.2 MIPI Interface

UD65640 MIPI DSI interface complies with the following standards:

MIPI DSI: Version 1.01.00r11 21-Feb-2008

MIPI D-PHY: Version 1.00.00 14-May-2009

MIPI DCS: Version 1.01.00

Basic DSI Specification

- Number of Lane: 2 lanes
- Maximum data rate: 500Mbps / Lane
- Both Video Mode and command mode supported

DSI System Configuration

Lane Pair	Host (Master) / UD65640 (Slave)
Clock Lane CLKP / CLKN	- Unidirectional Lane - Clock Only
Data Lane 0 (D0P/D0N)	- Bi-directional Lane - Forward High Speed - Bi-directional Escape Mode - Bi-directional LPDT
Data Lane 1 (D1P/D1N)	- Unidirectional Lane - Forward High Speed - NO LPDT

Table 3 - MIPI Interface Configuration

Pin Oder

Polarity and data lane swap are also supported with the following options configurable through PNSWAP, DSWAP1, and DSWAP0 pins.

Pin			MIPI Interface Pins					
PNSWAP	DSWAP1	DSWAP0	D0P	D0N	CKP	CKN	D1P	D1N
0	0	0	D0P	D0N	CKP	CKN	D1P	D1N
1	0	0	D0N	D0P	CKN	CKP	D1N	D1P
0	0	1	D1P	D1N	CKP	CKN	D0P	D0N
1	0	1	D1N	D1P	CKN	CKP	D0N	D0P

Table 4 - MIPI Pin Oder

Lane State Definition

MIPI DPHY uses data and clock lane differential pairs for DSI. Both differential lane pairs can be driven Low Power (LP) or High Speed (HS) mode. Low Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disable (A termination resistor of the receiver is disable) and it can be driven into a low power mode. High Speed mode means that differential pairs (The termination resistor of the receiver is enabled) are not used in the single end mode. There are used different modes and protocol in each mode when there is wanted to transfer information from the HOST to the driver IC and vice versa.

State code	Line voltage levels		High speed	Low power	
	Dp-line	Dn-line	Burst mode	Control mode	Escape mode
HS-0	HS Low	HS High	Differential-0	1	1
HS-1	HS High	HS Low	Differential-1	1	1
LP-00	LP Low	LP Low	N/A	Bridge	Space
LP-01	LP Low	LP High	N/A	HS-Rqst	Mark-0
LP-10	LP High	LP Low	N/A	LP-Rqst	Mark-1
LP-11	LP High	LP High	N/A	Stop	2

Table 5 - Lane State Descriptions

- Notes: 1. During high-speed transmission, the low power receivers observe LP-00 on the lines.
2. If LP-11 occurs during Escape mode, the lane returns to Stop state (Control mode LP-11).

CLK Lane

Clock lane state diagram is shown as following.

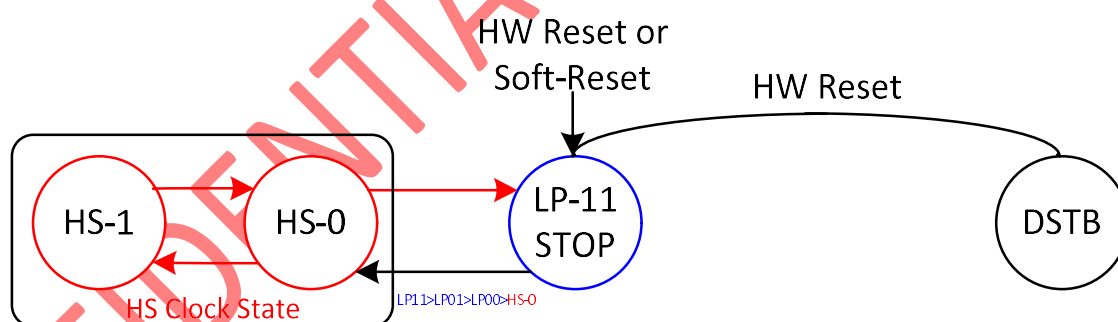


Figure 7 -Clock Lane State Diagram

Data Lane 0

Data Lane 0 supports bi-directional communication between host and client, in other words, only data lane 0 could support bus turn around request in UD65640, for others, only host -> client transmissions are allowed. Data Lane 0 can be driven in different modes, including Escape Mode, High-Speed Data Transmission and Bus Turnaround Request. The following is the data lane 0 state diagram.

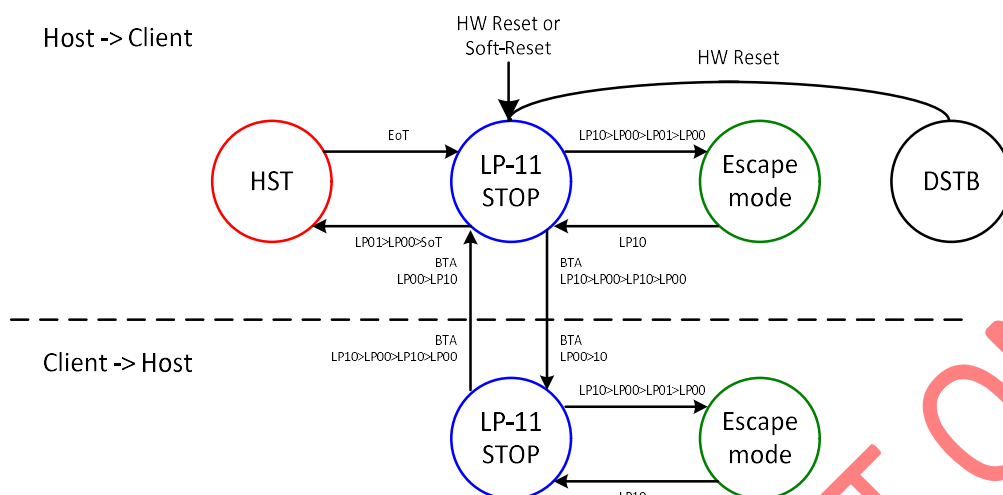


Figure 8 - Data Lane 0 State Diagram

No.	Description	Operation code	Note
1	High Speed Data Transmission	LP-11 > LP-01 > LP-00	
2	Escape mode entry	LP-11 > LP-10 > LP-00 > LP-01 > LP-00	
3	Turnaround	LP-11 > LP-10 > LP-00 > LP-10 > LP-00	
4	Exit Escape mode (Mark-1)	LP-10	
5	Deep Standby Mode	Command set_dstb 0x4F	1
6	Exit Deep Standby Mode	HW reset	2

Table 6 - Data Lane 0 Operating Modes

- Note:
1. command must be sent by Escape mode in Sleep mode.
 2. After exiting from the Deep Standby Mode, all of commands are reset.

Escape Mode Support

Escape Command	Command Type	Command (First bit -> Last bit)	implementation		Note
			LP-RX	LP-TX	
Low Power Data Transmission (LPDT)	Mode	1110_0001	Yes	Yes	
Ultra-Low Power State (ULPS)	Mode	0001_1110	Yes	No	
Remote Application Reset (RAR)	Trigger	0110_0010	No	No	1
TE-Report (TER)	Trigger	0101_1101	No	No	
Acknowledge (ACK)	Trigger	0010_0001	No	Yes	

Table 7 - Escape Mode Support

Note 1: DSI circuit is reset by Remote Application reset.

Packet Level Communication

1) Short Packet (SPa) Format

Short packet is used for most commands and associated parameters, where its format includes an 8-bit Data ID followed by two bytes of data and an 8-bit ECC.

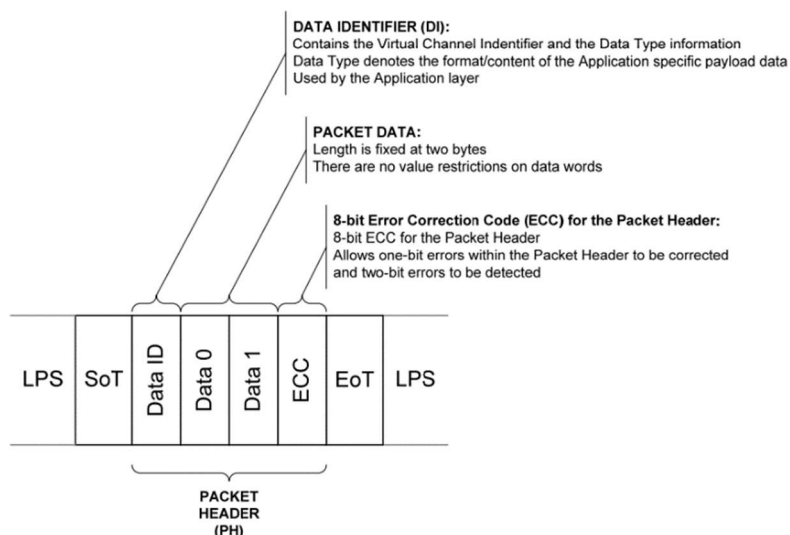


Figure 9 - Short Packet Format

2) Long Packet (LPa) Format

Long packets permit transmission of large blocks of pixel or data, which could range from 6 to 65,541 bytes in length.

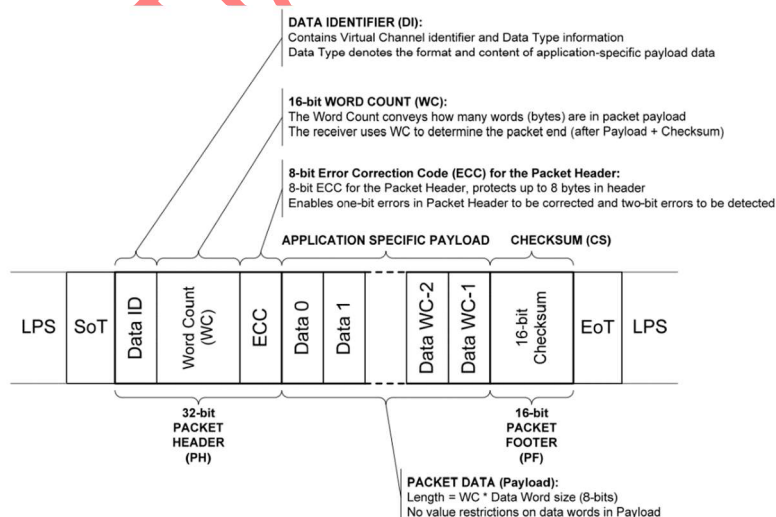


Figure 10 - Long Packet Format

7. Electrical Characteristics

7.1 Absolute Maximum Ratings

The absolute maximum rating is listed on the below table. When UD65640 product is used beyond the absolute maximum ratings, it may be permanently damaged.

Item	Symbol	Value	Unit	Notes
Power Supply Voltage (1)	IOVCC	TBD	V	
Power Supply Voltage (2)	AVCI18-AVSS	TBD	V	
	AVCI33-AVSS	TBD	V	
Power Supply Voltage (3)	ELVDD-AVSS	TBD	V	
Power Supply Voltage (4)	AVSS-ELVSS	TBD	V	
MIPI Differential Input	MIPI input	TBD	V	1
Input Voltage of Interface	V _{in}	TBD	V	
Output Voltage of Interface	V _o	TBD	V	
Operating temperature	T _{opr}	TBD	°C	
Storage temperature	T _{stg}	TBD	°C	

Table 8 - absolute maximum rating

Note: 1. MIPI input pins: CLKP/N, D0P/N, D1P/N

7.2 DC Characteristic

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit	Note
Power & Operation Voltage							
Analog Power Supply Voltage	AVCI18	Operating Voltage	TBD	1.8	TBD	V	
Analog Power Supply Voltage	AVCI33	Operating Voltage	TBD	3.3	TBD	V	
Digital I/O Power Supply (non-MIPI I/O)	IOVCC(1)	I/O supply voltage	TBD	1.8	TBD	V	
Digital I/O Power Supply (non-MIPI I/O)	IOVCC(2)	I/O supply voltage	TBD	3.3	TBD	V	
Digital I/O Input Level @Logic High	V _{IH}	IOVCC(1)(2)	0.7*IOVCC	-	IOVCC	V	
Digital I/O Input Level @Logic Low	V _{IL}	IOVCC(1)(2)	0	-	0.3*IOVCC	V	
Digital I/O Output Level @Logic High	V _{OH}		0.8*IOVCC	-	IOVCC	V	
Digital I/O Output Level @Logic Low	V _{OL}		0	-	0.2*IOVCC	V	
Digital I/O Input leakage @Logic High	IIHD	V _{in} = IOVCC			1	uA	
Digital I/O Input leakage @Logic Low	IILD	V _{in} = 0	-1			uA	
Logic High level input current (MIPI)	IIHD	V _{in} =0~DVDD			1	uA	
Logic Low level input current (MIPI)	IILD	V _{in} =0~DVDD	-1			uA	
ELVDD input Level	ELVDD	Operating Voltage		1.2		V	
ELVSS input Level	ELVSS	Operating Voltage	-8		-2	V	
VEMR Regulator voltage	VEMR		-6.8		-0.8	V	
AVEE Regulator voltage	AVEE		-6.8		-0.8	V	

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Table 9 - Voltage Setting Pattern Diagram

7.3 DSI DC/AC Characteristic

High speed receiver characteristic

Parameter	Description	Min	Typ.	Max	Unit
V _{CMRX(DC)}	Common-mode voltage HS receive mode	TBD	-	TBD	mV
ZID	Differential input impedance		100		Ω
V _{IDTH}	Differential input high threshold	-	-	TBD	mV
V _{IDTL}	Differential input low threshold	TBD	-	-	mV
V _{IHHS}	Single-ended input high voltage	-	-	TBD	mV
V _{ILHS}	Single-ended input low voltage	TBD	-	-	mV
CCM	Common-mode termination	-	-	2	pF

Table 10 – HS Receiver DC Specifications

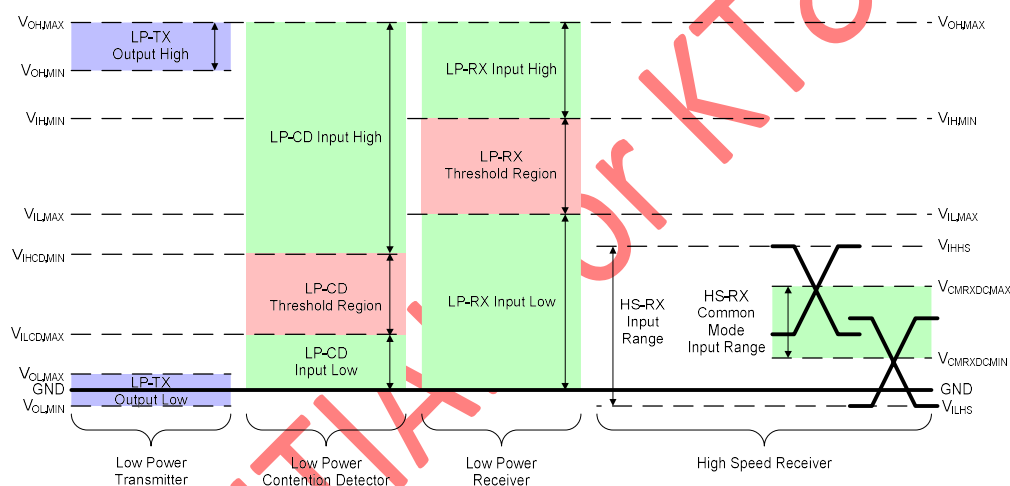


Figure 11 - D-PHY Signaling Levels

Low power receiver characteristic

Parameter	Description	Min	Typ.	Max	Unit
V _{IH}	Logic 1 input voltage	TBD	-	-	mV
V _{IL}	Logic 0 input voltage, not in ULP state	-	-	TBD	mV
V _{IL_ULPS}	Logic 0 input voltage, ULP state	-	-	TBD	mV
V _{HYST}	Input hysteresis	TBD	-	-	mV
eSPIKE	Input pulse rejection	-	-	TBD	V·ps
T _{MIN-RX}	Minimum pulse width response	TBD	-	-	

Table 11 - LP Receiver AC/DC specifications

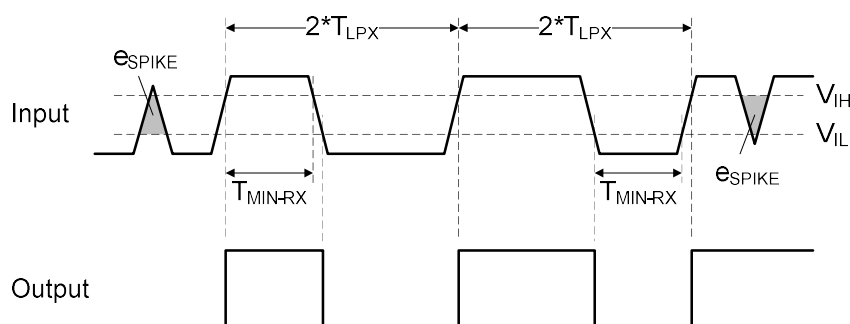


Figure 12 - Input Glitch Rejection of Low-Power Receivers

Low power Transmitter characteristic

Parameter	Description	Min	Typ.	Max	Unit
V _{OH}	The output high level	1.1	1.2	1.3	V
V _{OL}	The output low level	TBD	-	TBD	mV
ZOLP	Output impedance of LP transmitter	TBD	-	-	Ω
V _{IHCD}	Logic1 contention threshold	TBD	-	-	mV
V _{ILCD}	Logic0 contention threshold	-	-	TBD	mV

Table 12 - LP Transmitter DC Specifications

7.4 Timing Characteristics

7.4.1 High Speed Mode Characteristics

Parameter	Symbol	Min	Typ.	Max	Unit
UI instantaneous	U _{IINST}	TBD	-	TBD	ns
T Data to Clock Skew	T _{SKEW}	TBD	-	TBD	UIHS
RX Data to Clock Setup Time Tolerance	T _{SETUP}	TBD	-	-	UIHS
RX Data to Clock Hold Time Tolerance	T _{HOLD}	TBD	-	-	UIHS

Table 13 - Data-Clock Timing Specifications

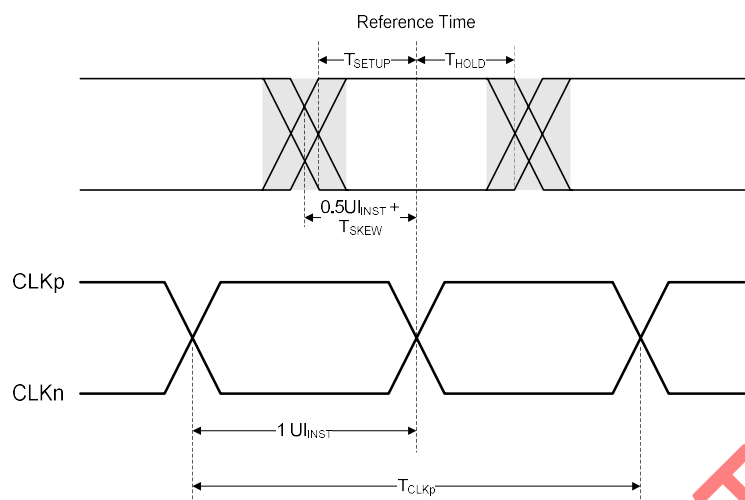


Figure 13 - Data to Clock Timing Definitions

7.4.2 Low Power Mode Characteristics

Parameter	Description	Min	Typ.	Max	Unit
T_{LPX}	Transmitted length of any Low-Power state period	TBD	-	-	ns
$T_{TA-SURE}$	Time that the new transmitter waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround	T_{LPX}		$2 * T_{LPX}$	
T_{TA-GET}	Time that the new transmitter drives the Bridge state (LP-00) after accepting control during a Link Turnaround	$5 * T_{LPX}$			
T_{TA-GO}	Time that the transmitter drives the Bridge state (LP-00) before releasing control during a Link Turnaround	$4 * T_{LPX}$			
T_{WAKEUP}	Time that a transmitter drives a Mark-1 state prior to a Stop state in order to initiate an exit from ULPS.	TBD			ms

Table 14 – LP TX Operation Timing Parameters

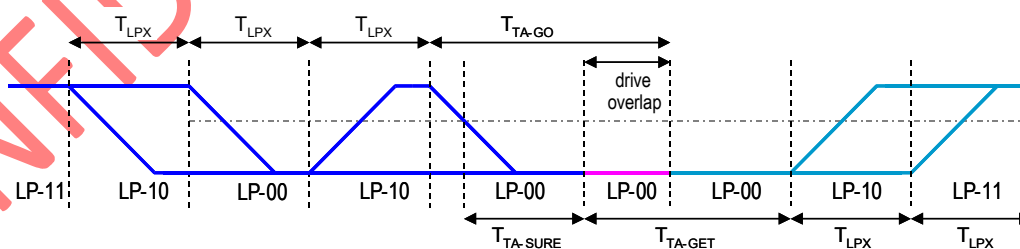


Figure 14 - Turnaround Procedure

7.4.3 High Speed Mode Operation Timing Characteristics

Table 15 - Global Operation Timing Parameters

Parameter	Description	Min	Typ.	Max	Unit	Notes
$T_{HS-PREPARE}$	Time to drive LP-00 to prepare for HS transmission	TBD	-	TBD	ns	
$T_{HS-PREPARE} + T_{HS-ZERO}$	$T_{HS-PREPARE}$ + Time to drive HS-0 before the Sync sequence	TBD	-	-	ns	
$T_{HS-TRAIL}$	Time to drive flipped differential state after last payload data bit of a HS transmission burst	TBD	-	-	ns	1,2
$T_{HS-EXIT}$	Time to drive LP-11 after HS burst	TBD	-	-	ns	
$T_{CLK-POST}$	Time that the transmitter shall continue sending HS clock after the last associated Data Lane has transitioned to LP mode	TBD	-	-	UI	3
$T_{CLK-PREPARE} + T_{CLK-ZERO}$	$T_{CLK-PREPARE}$ + time for lead HS-0 drive period before starting Clock	TBD	-	-	ns	
$T_{CLK-PRE}$	Time that the HS clock shall be driven prior to any associated Data Lane beginning the transition from LP to HS mode	TBD	-	-	UI	
$T_{CLK-PREPARE}$	Time to drive LP-00 to prepare for HS clock transmission	TBD	-	TBD	ns	
$T_{CLK-TRAIL}$	Time to drive HS differential state after last payload clock bit of an HS transmission burst	TBD	-	-	ns	
T_{EOT}	Time from start of $T_{HS-TRAIL}$ period to start of LP-11 state	-	-	TBD		2

- Notes:
- If $a > b$ then $\max(a, b) = a$, otherwise $\max(a, b) = b$
 - Where $n = 1$ for Forward-direction HS mode.
 - The Device can work with this specification although the end part of internal process is remained when Clock Lane enter LP-11 and delay but the Device can work without the remained process if $t_{CLK-POST}$ is more than 256 UI.

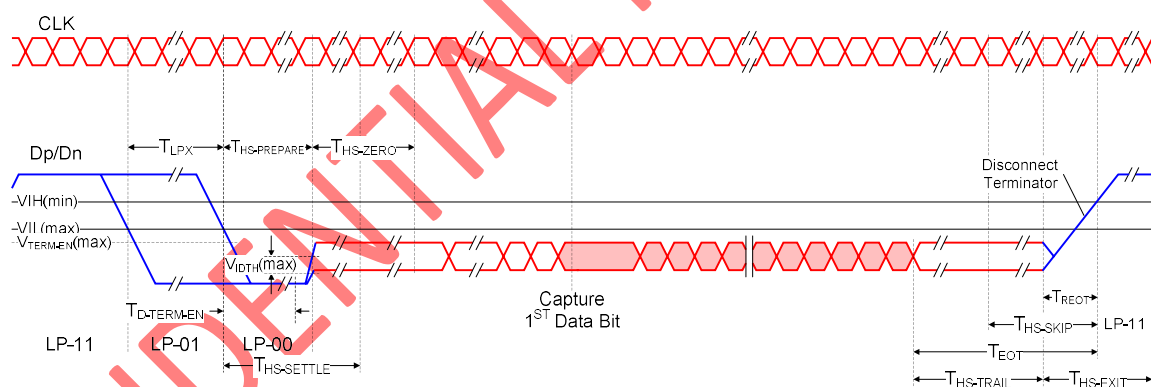


Figure 15 - High-Speed Data Transmission in Bursts

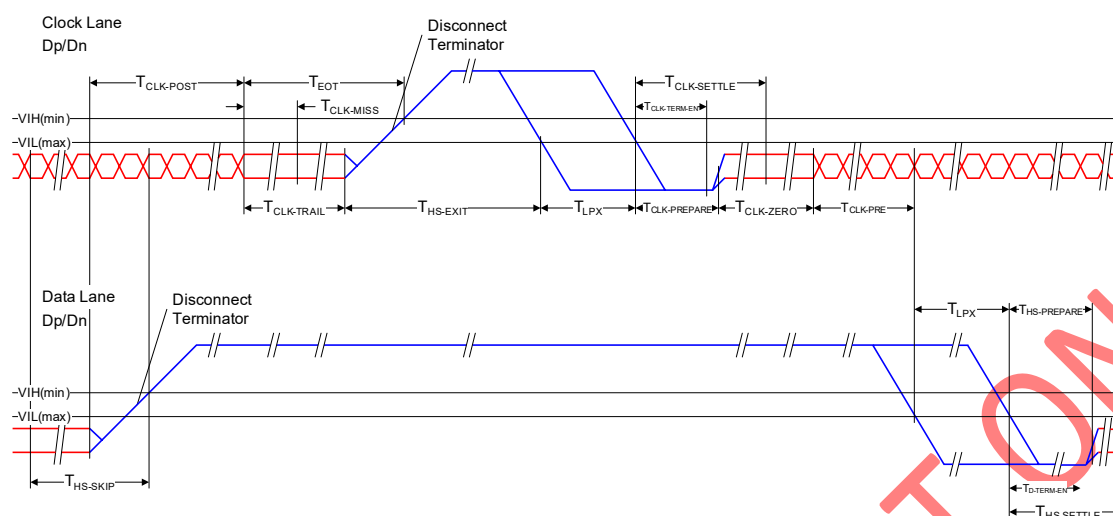


Figure 16 - Switching the Clock Lane between Clock Transmission and Low-Power Mode

7.5 Reset Timing Characteristics

When reset happens in Sleep-out mode, this UD65640 product will enter blanking sequence with the maximum time 120 msec. Then this UD65640 product will remain in blanking state and return \ default state. During reset complete time (t_{RT}), data in OTP will be re-loaded and latched to internal registers. This data re-load is done every time when there is an H/W reset and completes within 20 msec after the rising edge of RESX.

Therefore, it is necessary to wait at least 20 msec after releasing the RESX before sending commands. Moreover, the Sleep-out command cannot be sent in 120 msec. Spike (less than 20ns width) Rejection can also be applied during a valid reset pulse.

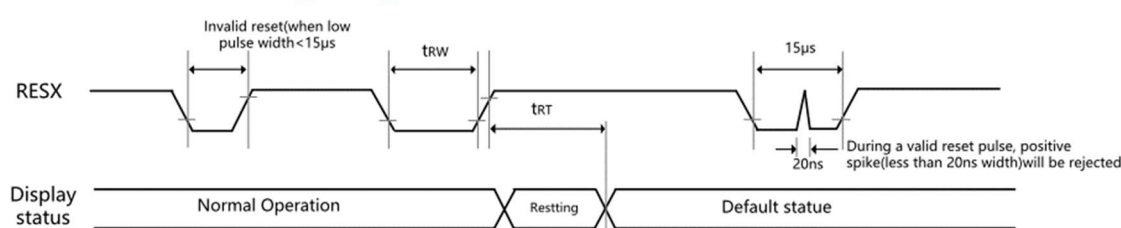


Figure 17 – RESX Timing

Single	Symbol	Parameter	Min	Type.	Max.	Unit	Description
RESX	tRW	Reset low pulse width	15			us	
	tRT	Reset complete time				ms	When reset applied at sleep-in mode
						ms	When reset applied at sleep-out mode

Table 16 – RESX Timing Parameters

7.6 Power Consumption

TBD

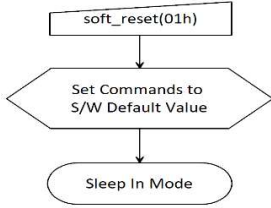
8. User Command

8.1 Table User Command

Operational Code (Hex)	Command	Command ® /Read ®/Write(W)	Number Of Parameter	Command Accessibility		Note
				Sleep Mode Off	Sleep Mode On	
01h	soft_reset	C	0	Yes	Yes	
0Ah	get_power_mode	R	1	Yes	Yes	
0Bh	get_address_mode	R	1	Yes	Yes	
0Eh	get_signal_mode	R	1	Yes	Yes	
10h	enter_sleep_mode	C	0	Yes	Yes	
11h	exit_sleep_mode	C	0	Yes	Yes	
28h	set_display_off	C	0	Yes	Yes	
29h	set_display_on	C	0	Yes	Yes	
2Ch	write_memory_start	W	N	Yes	Yes	
34h	set_tear_off	C	0	Yes	Yes	
35h	set_tear_on	W	1	Yes	Yes	
36h	set_address_mode	W	1	Yes	Yes	
3Ch	wite_memory_continue	W	N	Yes	Yes	
44h	set_tear_scanline	W	2	Yes	Yes	
45h	get_scanline	R	2	Yes	Yes	
4Fh	set_dstb	C	0	No	Yes	
51h	set_display_brightness	W	1	Yes	Yes	
52h	get_display_brightness	R	1	Yes	Yes	
DAh	Read_ID1	R	1	Yes	Yes	
DBh	Read_ID2	R	1	Yes	Yes	
DCh	Read_ID3	R	1	Yes	Yes	

Table 17 - User Command Accessibility

soft_reset: 01h

0Ah	soft_reset									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H →D	0	0	0	0	0	0	0	1	01h
Description	When the Software reset command is written, it causes software reset. It resets the commands and parameters to their S/W reset default values. (See default tables in each command description.) ※Sof reset Command cannot be sent during Sleep Out sequence.									
Flow Chart	 <pre> graph TD A[soft_reset(01h)] --> B{{Set Commands to S/W Default Value}} B --> C([Sleep In Mode]) </pre>									

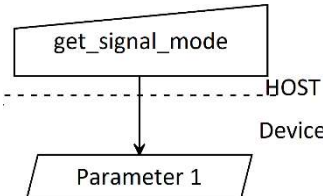
get_power_mode: 0Ah

0Ah	get_power_mode										
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex	
Command	H → D	0	0	0	0	1	0	1	0	0Ah	
Parameter 1	D → H	0	0	0	SLPOUT	NORON	DSPON	0	1	01h	
Description	The display module returns the current power mode as listed below.										
	Bit	Description			Comment	Command list symbol					
	B [7:5]	Reserved									
	B4	Sleep Mode On/Off				SLPOUT					
	B3	Display Normal Mode On/Off				NRMLMODE					
	B2	Display On/Off				DSPON					
	B [1:0]	Reserved									
	• B [7:5] –Reserved; This bit is not applicable. • B4 – Sleep Mode On/Off; ‘1’ = Sleep Mode On, ‘0’ = Sleep Mode Off. • B3 – Display Normal Mode On/Off; ‘1’ = Normal Mode On, ‘0’ = Normal Mode Off. • B2 – Display On/Off, ‘0’ = Display is Off, ‘1’ = Display is On. • B [1:0] – Reserved; This bit is not applicable.										
	Flow Chart	get_power_mode(0Ah) HOST Device Parameter 1									

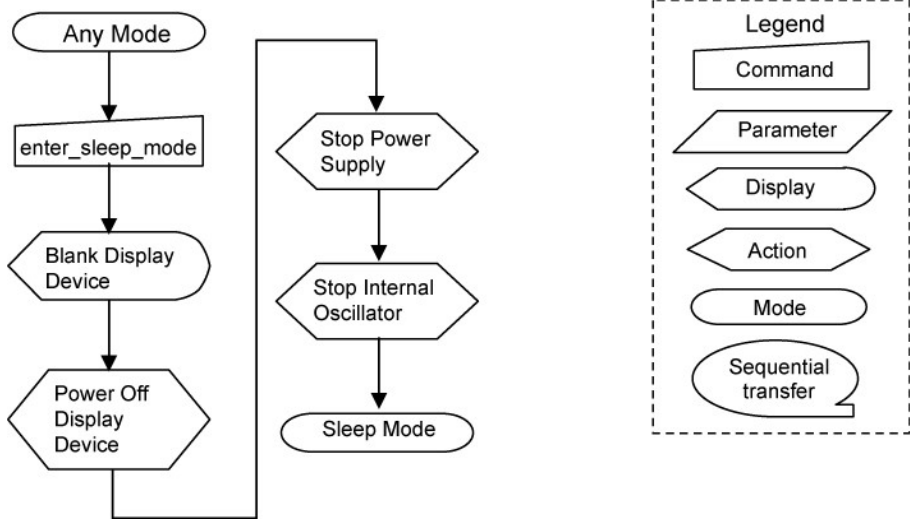
get_address_mode: 0Bh

0Bh	get_address_mode										
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex	
Command	H →D	0	0	0	0	1	0	1	1	0Bh	
Parameter 1	D → H	0	0	0	0	RGB	0	FH	0	00h	
Description	This command indicates the current status of the display as described in the table below:										
	Bit	Description				Comment					
	B[7:4]	Reserved									
	B3	RGB				RGB/BGR Order					
	B2	Reserved									
	B1	FH				Flip Horizontal					
	B0	Reserved									
	• B [7:4] –Reserved; This bit is not applicable. • B3 –‘0’ = RGB, ‘1’ = BGR • B2 – Reserved; This bit is not applicable. • B1 –This bit flips the image shown on the display device left to right. No change is made to the frame memory. ‘0’ = Normal , ‘1’ = Flipped • B0 – Reserved; This bit is not applicable.										
	Flow Chart										
		get_address_mode(0Bh) HOST Device Parameter 1									

get_signal_mode: 0Eh

0Eh	get_signal_mode									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	0	0	1	1	1	0	0Eh
Parameter 1	D → H	TEON	TEMODE	0	0	0	0	0	0	00h
Description	The display module returns the display signal mode. - Bit D7 – Tearing effect line on/off; ‘1’ = On, ‘0’ = Off. - Bit D6 – Tearing effect line mode; ‘1’ = mode 2, ‘0’ = mode 1. - Bit D0 – DSI Error; ‘1’ = error detected, ‘0’ = no errors. - X = Don’t care									
Restriction										
Flow chart	 <pre> graph TD Host[HOST] -- "get_signal_mode" --> Device[Device] Device -- "Parameter 1" --> Param1[/Parameter 1/] </pre>									

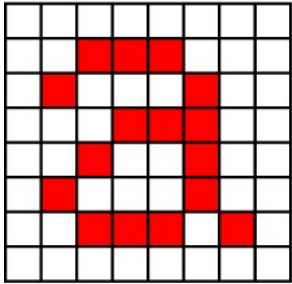
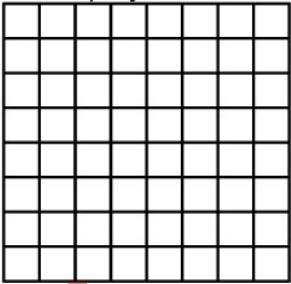
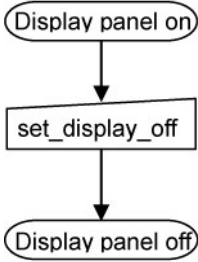
enter_sleep_mode: 10h

10h	enter_sleep_mode									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	0	1	0	0	0	0	10h
Parameter	None									
Description	This command causes the display module to enter the sleep mode. In this mode, the step-up circuit, internal oscillator and panel scanning stop. See State Transition Diagram for each stage of transition. X = Don't care									
Restriction	This command has no effect when the module is already in Sleep mode. Sleep mode can be exited only when the exit_sleep_mode (11h) is received. Sending a new command is prohibited when the chip is in display off sequence or power off sequence.									
Flow chart										

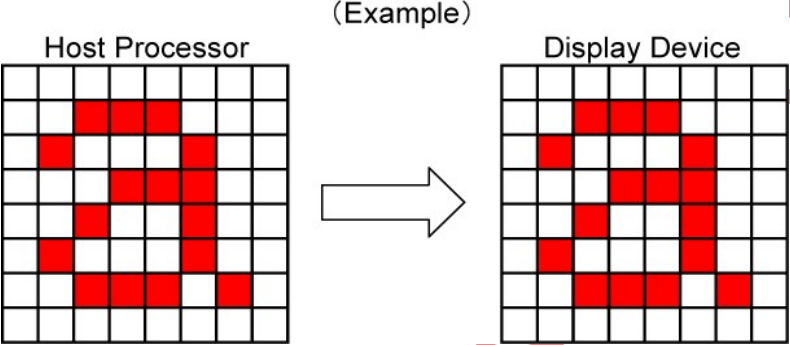
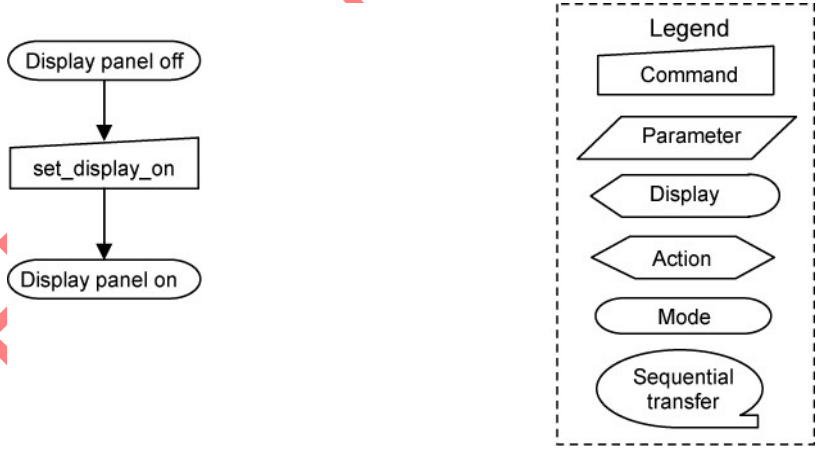
exit_sleep_mode: 11h

11h	exit_sleep_mode									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	0	1	0	0	0	1	11h
Parameter	None									
Description	This command causes the display module to exit Sleep mode. The step-up circuit, internal oscillator, and panel scanning start. See State Transition Diagram for each stage of transition. X = Don't care									
Restriction	This command shall not cause any visual effect on display device when the display module is not in Sleep mode. No new command setting is allowed during power on sequence. Operation may continue for more than 6 frame periods due to power on sequence setting. Do not send any command either in this case. The host processor must wait 5 frame periods after sending an enter_sleep_mode command before sending an exit_sleep_mode command.									
Flow chart	<pre> graph TD SM1([Sleep Mode]) --> ESM[exit_sleep_mode] ESM --> SIO{{Start Internal Oscillator}} SIO --> SPS{{Start Power Supply}} SPS --> SM2([Sleep Mode]) PODD{{Power On Display Device}} --> BDD([Blank Display Device]) BDD --> DMC([Display Memory contents]) DMC --> SM3([Sleep Mode]) subgraph Legend C[Command] P[/Parameter/] D{{Display}} A{{Action}} M([Mode]) ST([Sequential transfer]) end </pre>									

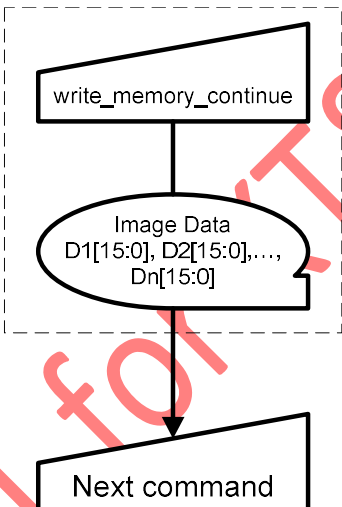
set_display_off: 28h

28h	set_display_off									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	1	0	1	0	0	0	28h
Parameter	None									
Description	This command causes the display module to stop displaying image data on the display device. No status bits are changed. (Example) Host Processor  → Display Device  X = Don't care									
Restriction	This command has no effect when the display panel is already off.									
Flow chart	 Legend - Command - Parameter - Display - Action - Mode - Sequential transfer									

set_display_on: 29h

29h	set_display_on									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	1	0	1	0	0	1	29h
Parameter	None									
Description	This command causes the display module to start displaying the image data on the display device. No status bits are changed. (Example)  X = Don't care									
Restriction	This command has no effect when the display panel is already on.									
Flow chart										

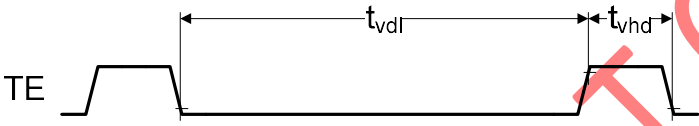
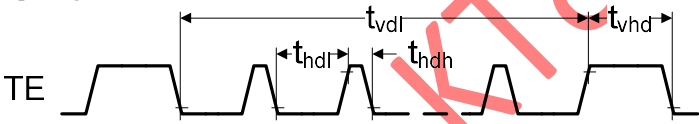
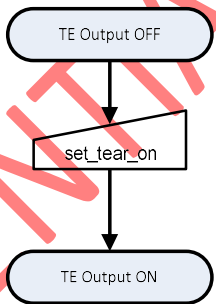
write_memory_start: 2Ch

2Ch	write_memory_start									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	1	0	1	1	0	0	2Ch
Parameter 1	H → D	Pixel1								
Parameter 2	H → D	Pixel2								
...	H → D	...								
Parameter N	H → D	Pixel N								
Description	This command transfers image data from the host processor to the display module's frame memory starting at the pixel location specified.									
Restriction										
Flow chart	 <pre> graph TD A[/write_memory_continue/] --> B([Image Data D1[15:0], D2[15:0], ..., Dn[15:0]]) B --> C[/Next command/] subgraph DashedBox [] A B end </pre>									

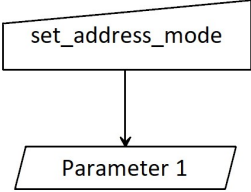
set_tear_off: 34h

34h	set_tear_off									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H →D	0	0	1	1	0	1	0	0	34h
Parameter	None									
Description	This command turns off the display module's Tearing Effect output signal from the TE signal line. X = Don't care									
Restriction	This command has no effect when the Tearing Effect output is already off.									
Flow chart	<pre> graph TD A([TE Output ON]) --> B[/set_tear_off/] B --> C([TE Output OFF]) </pre> Legend - Command: Trapezoid - Parameter: Parallelogram - Display: Rounded rectangle - Action: Pointed rectangle - Mode: Rounded rectangle - Sequential transfer: Oval with tail									

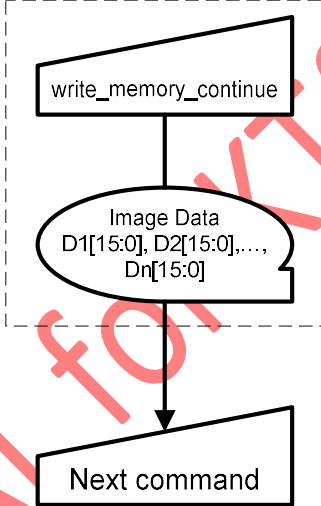
set_tear_on: 35h

35h	set_tear_on									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	1	1	0	1	0	1	35h
Parameter 1	H → D	0	0	0	0	0	0	0	TELOM	00h
Description	This command turns on the display module's Tearing Effect output signal from the TE signal line. Two modes are supported, If TELOM=0, the TE output signal follows V-SYNC information.  If TELOM=1, the TE output signal follows H-SYNC information.  The Tearing Effect output signal line shall be in LOW state when the display module is in Sleep mode.									
Restriction	This command has no effect when the Tearing Effect output is already ON.									
Flow chart	 Legend - Command - Parameter - Display - Action - Mode - Sequential transfer									

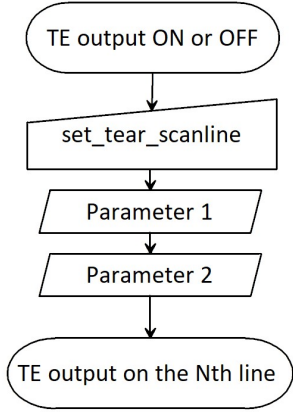
set_address_mode: 36h

36h	set_address_mode									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H →D	0	0	1	1	0	1	1	0	36h
Parameter 1	H →D	0	0	0	0	RGB	0	FH	FV	00h
Description	This command sets scanning direction of data transferred from the host. - B3 – RGB/BGR Order; ‘0’ = RGB, ‘1’ = BGR. The figure below shows relationships between input data and display data. For details, see “Relationship between Input Data and Display Data.” - B1 – Horizontal Flip; ‘0’ = Normal display, ‘1’ = Flipped display. This bit flips the image shown on the display device left to right. No change is made to the frame memory.									
Restriction										
Flow chart	 <pre> graph TD A[set_address_mode] --> B[/Parameter 1/] </pre>									

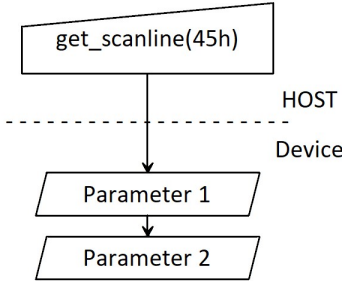
write_memory_continue: 3Ch

3Ch	write_memory_continue									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	0	1	1	1	1	0	0	3Ch
Parameter 1	H → D	Pixel 1								
Parameter 2	H → D	Pixel 2								
...	...	...								
Parameter N	H → D	Pixel N								
Description	This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location.									
Restriction										
Flow chart	 <pre> graph TD A[write_memory_continue] --> B([Image Data D1[15:0], D2[15:0], ..., Dn[15:0]]) B --> C[Next command] subgraph Box [] A B end </pre>									

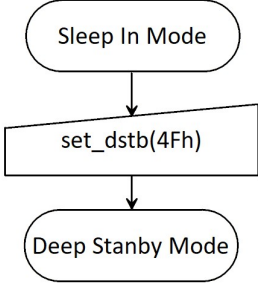
set_tear_scanline: 44h

44h	set_tear_scanline									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	1	0	0	0	1	0	0	44h
Parameter 1	H → D	0	0	0	0	0	0	N[9]	N[8]	00h
Parameter 2	H → D	N[7]	N[6]	N[5]	N[4]	N[3]	N[2]	N[1]	N[0]	00h
Description	This command turns on the display Tearing Effect output signal on the TE signal line when the display module reaches line N[9:0]. The Tearing Effect Line On has one parameter that describes the Tearing Effect Output Line mode.									
Restriction										
Flow chart	 <pre> graph TD A([TE output ON or OFF]) --> B[set_tear_scanline] B --> C[/Parameter 1/] C --> D[/Parameter 2/] D --> E([TE output on the Nth line]) </pre>									

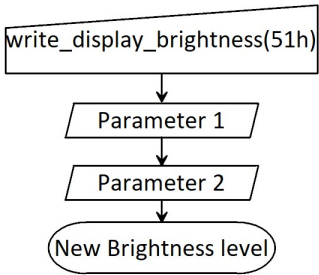
get_scanline: 45h

45h	get_scanline									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	1	0	0	0	1	0	1	45h
Parameter 1	D → H	0	0	0	0	0	0	N[9]	N[8]	00h
Parameter 2	D → H	N[7]	N[6]	N[5]	N[4]	N[3]	N[2]	N[1]	N[0]	00h
Description	The display returns the current scan line, N, used to update the display device. The total number of scan lines on a display device is defined as VSYNC + VBP + VACT + VFP. The first scan line is defined as the first line of V-Sync and is denoted as Line 0. When in Sleep Mode, the value returned by “get_scanline” is undefined.									
Restriction										
Flow chart	 <pre> graph TD A[get_scanline(45h)] --> B[Parameter 1] B --> C[Parameter 2] style A fill:#fff,stroke:#000 style B fill:#fff,stroke:#000 style C fill:#fff,stroke:#000 </pre>									

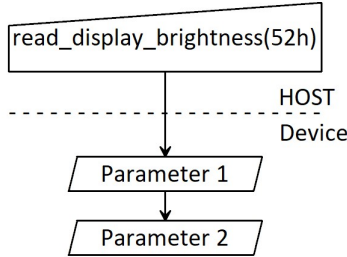
set_dstb: 4Fh

4Fh	set_dstb									
	Direction	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	H → D	0	1	0	0	1	1	1	1	4Fh
Description	This command is used to enter deep standby mode. Notes: 1. To exit Deep Standby Mode, set RESX low pulse. 2. If user wants to enter DSTB mode from Normal Display directly, it shall enter sleep-in & display-off mode first, and wait 8 frames or more time for completing power-down sequence, and then execute this command to enter DSTB mode.									
Restriction										
Flow chart	 <pre> graph TD A([Sleep In Mode]) --> B[set_dstb(4Fh)] B --> C([Deep Stanby Mode]) </pre>									

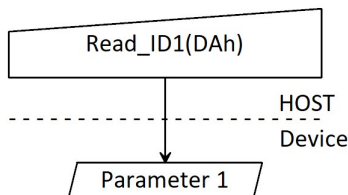
write_display_brightness: 51h

51h	write_display_brightness									
	Direction	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Hex
Command	H → D	0	1	0	1	0	0	0	1	51h
Parameter 1	H → D	DBV [7]	DBV [6]	DBV [5]	DBV [4]	DBV [3]	DBV [2]	DBV [1]	DBV [0]	00h
Parameter 2	H → D	0	0	0	0	0	0	DBV [9]	DBV [8]	00h
description	This command is used to adjust the display brightness . X = Don't care									
Restriction										
Flow chart	 <pre> graph TD A[write_display_brightness(51h)] --> B[/Parameter 1/] B --> C[/Parameter 2/] C --> D([New Brightness level]) </pre>									

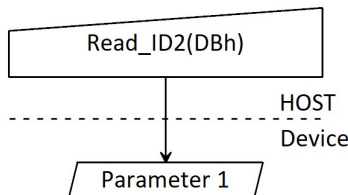
read_display_brightness: 52h

52h	read_display_brightness									
	Direction	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Hex
Command	H → D	0	1	0	1	0	0	1	0	52h
Parameter 1	D → H	DBV [7]	DBV [6]	DBV [5]	DBV [4]	DBV [3]	DBV [2]	DBV [1]	DBV [0]	80h
Parameter 2	D → H	0	0	0	0	0	0	DBV [9]	DBV [8]	00h
Description	This command returns the current brightness value. DBV[9:0] This register returns the brightness data of the display module that is currently used. X = Don't care									
Restriction										
Flow chart	 <pre> graph TD Command[read_display_brightness(52h)] --> Param1[/Parameter 1/] Param1 --> Param2[/Parameter 2/] </pre>									

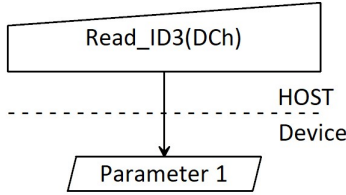
Read_ID1: DAh

DAh	Read_ID1									
	Direction	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Hex
Command	H → D	1	1	0	1	1	0	1	0	DAh
Parameter 1	D → H	DDB0 [7]	DDB0 [6]	DDB0 [5]	DDB0 [4]	DDB0 [3]	DDB0 [2]	DDB0 [1]	DDB0 [0]	00h
Description	This read byte identifies Module's manufacture Device Code DDB0									
Restriction										
Flow chart	 <pre> sequenceDiagram participant Host participant Device Host->>Device: Read_ID1(DAh) Device-->>Host: Parameter 1 </pre>									

Read_ID2: DBh

DBh	Read_ID2									
	Direction	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Hex
Command	H → D	1	1	0	1	1	0	1	1	DBh
Parameter 1	D → H	DDB1 [7]	DDB1 [6]	DDB1 [5]	DDB1 [4]	DDB1 [3]	DDB1 [2]	DDB1 [1]	DDB1 [0]	00h
Description	This read byte identifies Module's manufacture Device Code DDB1									
Restriction										
Flow chart	 <pre> sequenceDiagram participant Host participant Device Host->>Device: Read_ID2(DBh) Device-->>Host: Parameter 1 </pre>									

Read_ID3: DCh

DCh	Read_ID3									
	Direction	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Hex
Command	H → D	1	1	0	1	1	1	0	0	DCh
Parameter 1	D → H	DDB2 [7]	DDB2 [6]	DDB2 [5]	DDB2 [4]	DDB2 [3]	DDB2 [2]	DDB2 [1]	DDB2 [0]	00h
Description	This read byte identifies Module's manufacture Device Code DDB2									
Restriction										
Flow chart	 <pre> sequenceDiagram participant Host participant Device Host->>Device: Read_ID3(DCh) Device-->>Host: Parameter 1 </pre>									

9. Power Generation

9.1 Power Generation Scheme

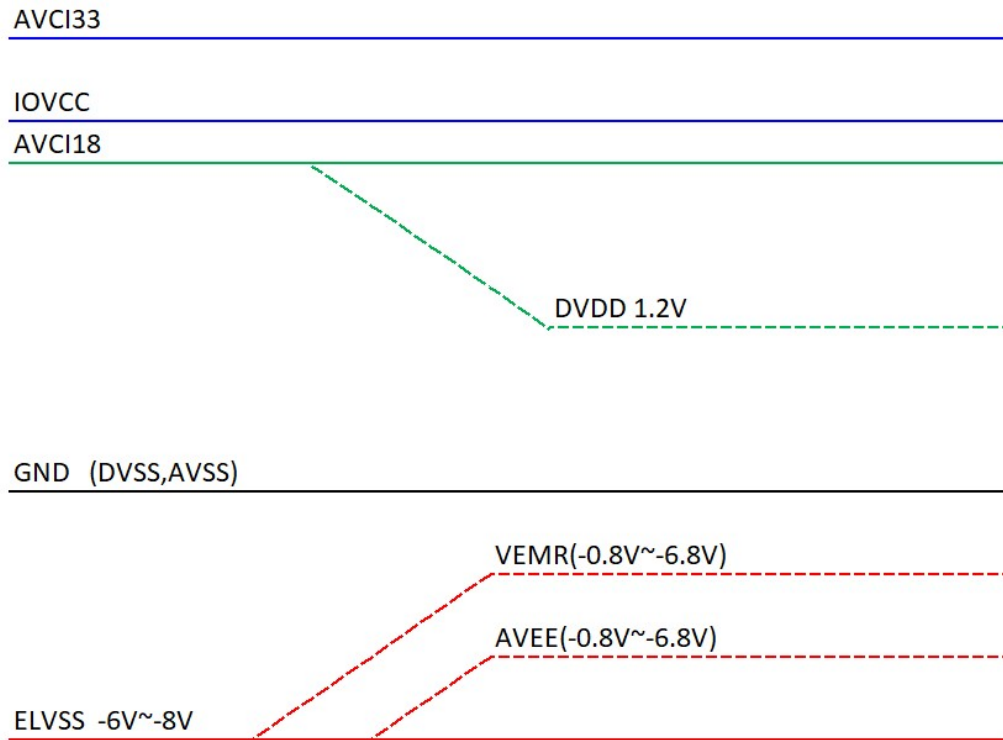


Figure 18 - Power Generation

9.2 Power Sequence

9.2.1 Power On Sequence

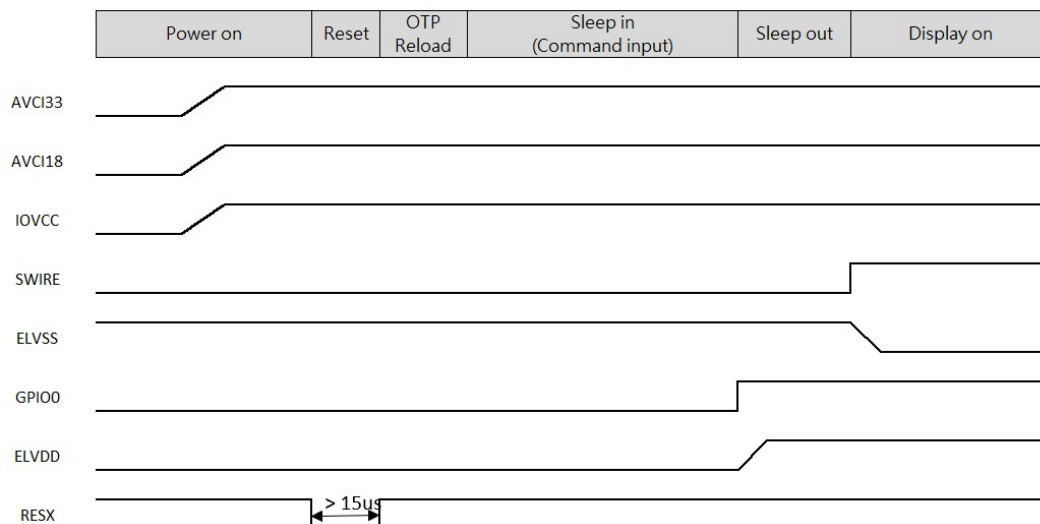


Figure 19 - Power On Sequence Timing

Note:

1. ELVSS power on timing is dependent on power IC, we suggest customer to select power IC that power on timing $\leq 4mS$.
2. GPIO0 is used to enable external ELVDD power IC, enable timing could be adjusted via register. It is suggested that external ELVDD power IC power on timing $\leq 4mS$.

9.2.2 Power Off Sequence

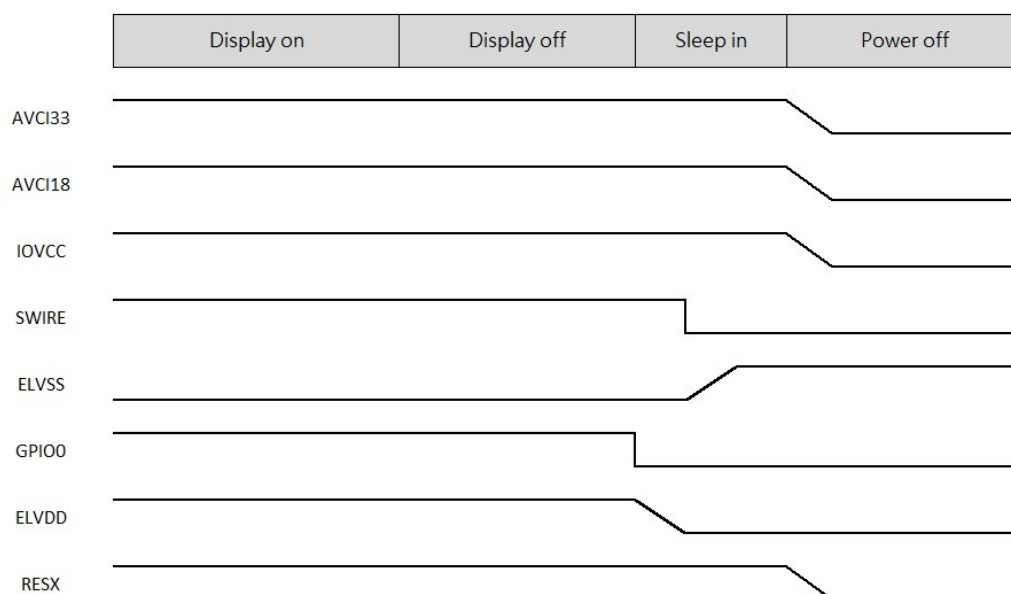


Figure 20 - Power Off Sequence Timing

10. Function Descriptions

10.1 Temperature Compensation Function

In general, the luminance of an OLED display is influenced by panel temperature variations. To address this issue, there are four sensors located around the panel that continuously monitoring backplane temperature. This feature allows for the adjustment of operating conditions to maintain a relatively constant luminance to deal with the environmental changes.

10.2 Temperature Protection Function

This function is designed to prevent damage to the backplane device when the device temperature exceeds the the designated protection threshold. When enabled, the protection function will trigger a notification (to CPU), and also turns off the display.

10.3 Orbit Function

UD65640 supports two resolutions: 640x480 and 640x400, the start position of both resolution data images can be changed.

The horizontal display starting position can be moved left or right, the horizontal display starting position changes by one value and moves by one pixel. The resolution variable range of both is ± 4 pixels.

The vertical display starting position can be moved up or down, the vertical display starting position changes by one value and moves by one pixel. The resolution variable range of both is ± 6 pixels.

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Figure 21 – Resolution 640x480 orbit diagram

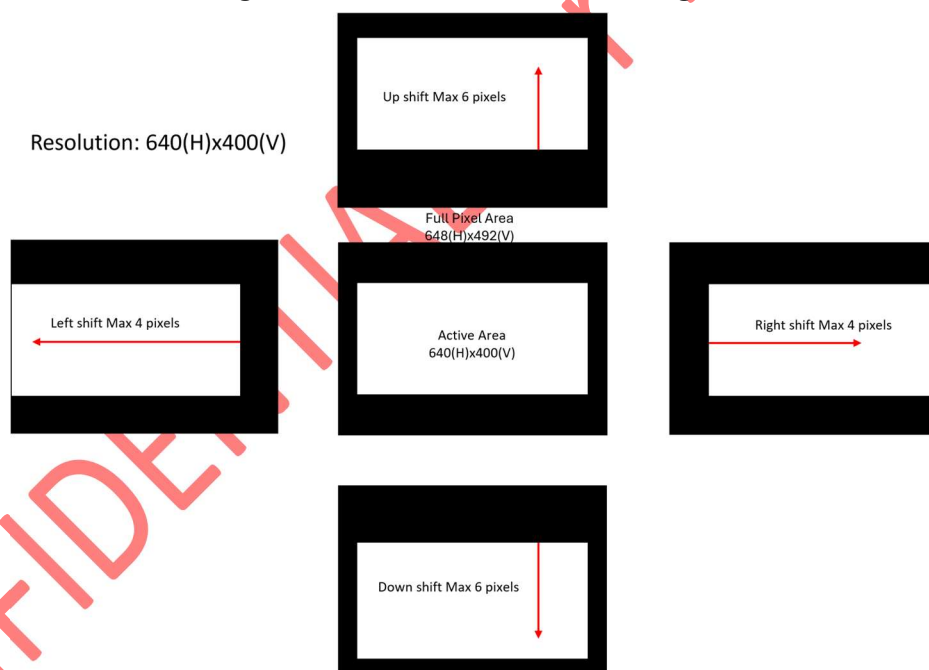


Figure 22 – Resolution 640x400 orbit diagram

10.4 Deburn-in Function

The deburn-in function can be configured to periodically make slight shifts in the displayed image, preventing persistent bright spots from remaining lit for extended periods and potentially causing permanent damage to the display.

10.5 Tearing Effect Output

The tearing effect output line supplies to the HOST a panel synchronization signal. This signal can be enabled or disabled by the `set_tear_off` (34h) and `set_tear_on` (35h) commands. The mode of the tearing effect signal is defined by the parameter of the `set_tear_on` (35h) and `set_tear_scanline`(44h) commands. The signal can be used by the HOST to synchronize internal VSYNC when displaying video images.

Tearing Effect Line Mode

Mode 1, the tearing effect output signal consists of V-sync information only:

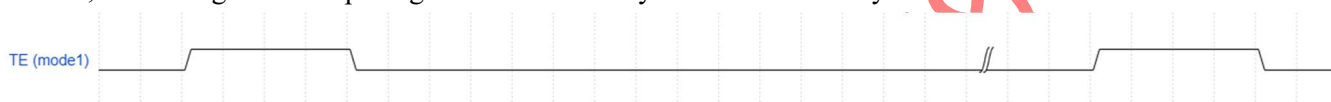


Figure 23 TE output V-SYNC

Mode 2, the tearing effect output signal consists of V-sync and H-sync information:

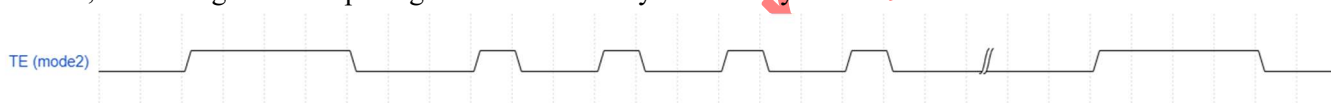


Figure 24 TE output H-SYNC

Mode 3, this mode turn on the tearing effect output signal when vertical scanning reaches line N.

N = The N-th scanning line which set by register N[9:0] of command STESL(44h).

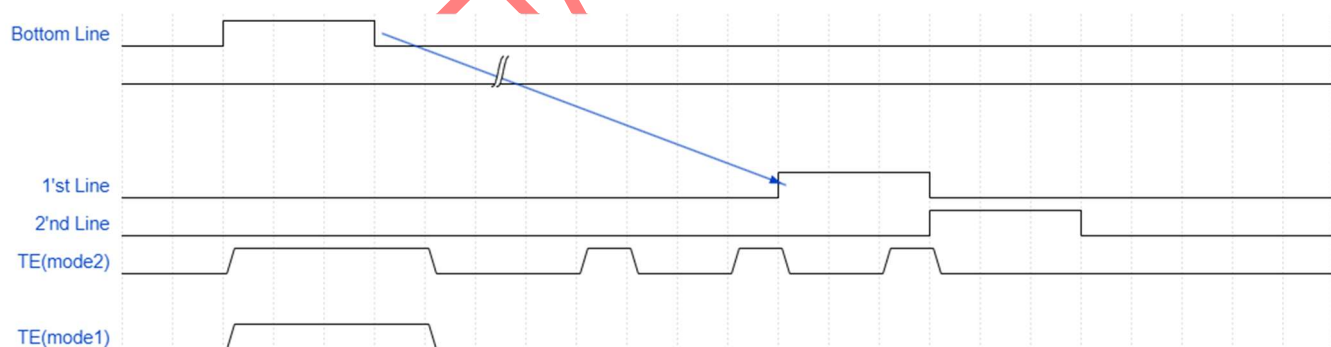


Figure 25 – Mode 3 vertical scanning reaches line

11. Wafer Map Diagram

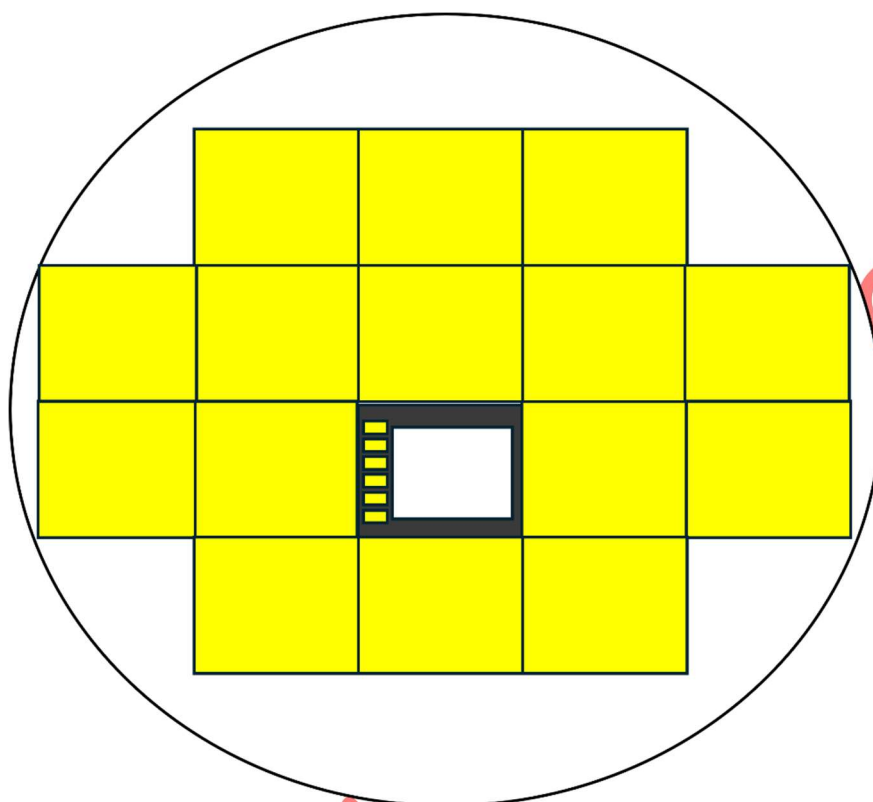


Figure 26 - Wafer Map Diagram

12. Chip Floorplan and Size

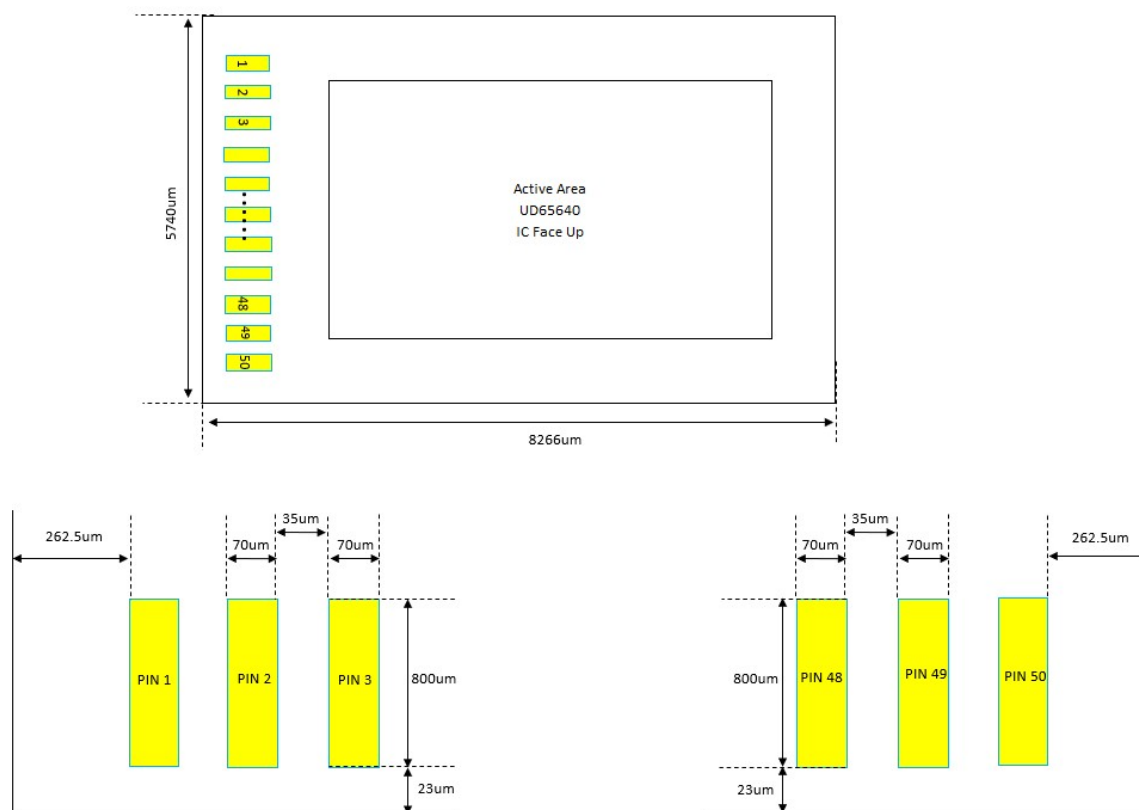


Figure 27 - Chip Outline and Size

13. Revision History

Version No.	Date	Page	Description
P0000	2024/04/15	50	Initial Version

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